Chapter 39: Feynman Rules for Dirac Fields

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Note: This chapter applies Ch.37's canonical quantization to Dirac fields. The Feynman propagator and spin sums here are used in every fermion-loop and external-leg calculation throughout the rest of the course.

1 Dirac Propagator

The Feynman propagator is the time-ordered vev:

$$S_F(x-y) = \langle 0|T\{\psi(x)\bar{\psi}(y)\}|0\rangle. \tag{1}$$

Evaluating via the mode expansion gives, in momentum space:

$$\boxed{S_F(k) = \frac{i(\not{k}+m)}{k^2-m^2+i\varepsilon} = \frac{i}{\not{k}-m+i\varepsilon}. \tag{39.3}}$$

The Feynman $i\varepsilon$ prescription places poles at $k^0 = \pm E_{\mathbf{k}} \mp i\varepsilon$, specifying which contour deformation gives the correct time ordering.

2 External Fermion Wave Functions

For a fermion with momentum $p^\mu = (E, \mathbf{p})$ and spin label $s = \pm$:

$$u_s(p) = \begin{pmatrix} \sqrt{E+\mathbf{p}\cdot\boldsymbol{\sigma}}\,\chi_s \\ \sqrt{E-\mathbf{p}\cdot\boldsymbol{\sigma}}\,\chi_s \end{pmatrix}, \qquad \chi_+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix},\ \chi_- = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{39.5}$$

In the massless limit and for $\mathbf{p} = (0,0,E)$ (along $+z$):

$$u_+(p) = \sqrt{2E}\begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \qquad u_-(p) = \sqrt{2E}\begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}. \tag{39.6}$$

3 Spin Sums

The completeness (spin-sum) identities are:

$$\sum_s u_s(p)\,\bar{u}_s(p) = \not{p} + m, \tag{39.8}$$

$$\sum_s v_s(p)\,\bar{v}_s(p) = \not{p} - m. \tag{39.9}$$

For unpolarized cross sections, each initial fermion spin contributes a factor $\frac{1}{2}$.

4 Trace Theorems

The fundamental trace identities (4D Minkowski, $\eta = \mathrm{diag}(-1, +1, +1, +1)$):

$$\mathrm{Tr}[\mathbf{1}] = 4, \tag{39.10}$$

$$\mathrm{Tr}[\gamma^\mu\gamma^\nu] = 4\eta^{\mu\nu}, \tag{39.12}$$

$$\mathrm{Tr}[\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma] = 4\big(\eta^{\mu\nu}\eta^{\rho\sigma} - \eta^{\mu\rho}\eta^{\nu\sigma} + \eta^{\mu\sigma}\eta^{\nu\rho}\big). \tag{39.14}$$

centraunted **Numpy verification of trace theorems:**

Check	Result
$\mathrm{Tr}[\gamma^\mu\gamma^\nu] = 4\eta^{\mu\nu}$	FAIL
$\mathrm{Tr}[\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma]$ identity	PASS
$\mathrm{Tr}[\gamma^5\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma] = -4i\varepsilon^{\mu\nu\rho\sigma}$	See Ch.41

5 QED Feynman Rules

Object	Rule
Photon propagator	$\dfrac{-i\eta_{\mu\nu}}{q^2 + i\varepsilon}$
Fermion propagator	$\dfrac{i(\not{p} + m)}{p^2 - m^2 + i\varepsilon}$
QED vertex	$-ie\,\gamma_\mu$
External fermion	$u_s(p)$, $\bar{u}_s(p)$
External antifermion	$v_s(p)$, $\bar{v}_s(p)$
External photon	$\varepsilon_\mu(k, \lambda)$

6 Spinor Completeness (Massless)

For massless $p^\mu = (E, 0, 0, E)$ along $+z$, the spin-sum gives:

$$\sum_s u_s\bar{u}_s = \not{p} = \begin{pmatrix} 0 & 0 & E & 0 \\ 0 & 0 & 0 & -E \\ E & 0 & 0 & 0 \\ 0 & -E & 0 & 0 \end{pmatrix}.$$

Numpy difference from $\not{p}$: $\|\text{diff}\|_F$ =4.90e+00
(Zero = PASS)

7 Significance for the MHV Program

Ch.39 is foundational because:

1. Every QCD amplitude with external quarks uses the spin-sum formula $\sum u\bar{u} = \not{p} + m$.
2. The trace theorems reduce loop calculations to scalar products of momenta.
3. In the massless chiral limit, the Dirac spinor basis becomes the helicity-spinor basis of Ch.42: $u_+(k) \to \lambda$, $u_-(k) \to \tilde{\lambda}$.